

Describing Data: Graphical and Numerical

Statistics I • Section 2

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Outline

- 2.1 Graphical methods
- 2.2 Measures of location
- 2.3 Measures of variability
- 2.4 Shape and association



Graphical Methods

Why describe, and how

- A **picture** shows the whole distribution at once: center, spread, shape, odd values.
- A **number** measures one feature exactly, so that we can compare and compute.
- A good description uses both. One statistic can hide the shape. A picture cannot be put into a formula.

Organizing the data first

Definition 2.1 (Frequency and relative frequency distribution)

- (i) A *frequency distribution* (次數分配) is a table of classes with the number of observations (the *frequency*) in each.
 - (ii) The *relative frequency* (相對次數) of a class is its frequency divided by the total number of observations n . Relative frequencies sum to 1.
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- Two totals to check every time: the frequencies add to n , the relative frequencies add to 1.

Example 2.2

Example 2.2

Twenty clients placed these numbers of trades last month:

7, 12, 2, 5, 16, 9, 10, 6, 14, 4, 8, 11, 15, 5, 13, 3, 7, 18, 10, 9.

Group them into 0–4, 5–9, 10–14, 15–19 and build the frequency and relative-frequency distribution. Check the two totals.

Which display, and why

- A frequency distribution is where a bar chart, pie chart or histogram begins.
- Other displays keep more: a time-series plot keeps the time order, a scatter plot keeps each pair (x_i, y_i) .
- So the graph depends on the type and the structure of the data, and on what we want to learn from it.

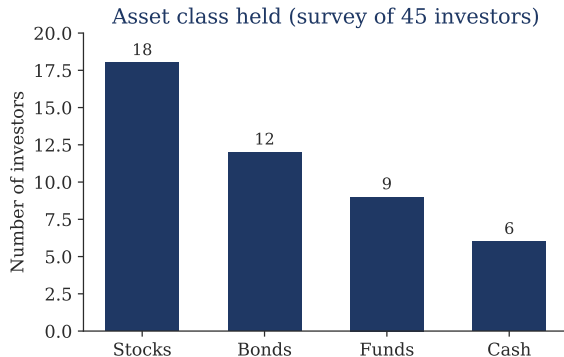
The bar chart

- **Bar chart** (長條圖): one bar per category, height = frequency. The bars are separated: the categories are distinct groups, not adjoining numerical intervals.

Example 2.3

A survey of 45 investors records the asset class each holds most: Stocks 18, Bonds 12, Funds 9, Cash 6. Display the counts.

Example 2.3: bar chart of asset class



The height of each bar is the category's frequency. The categories have no natural order, so we could rearrange them without losing anything.

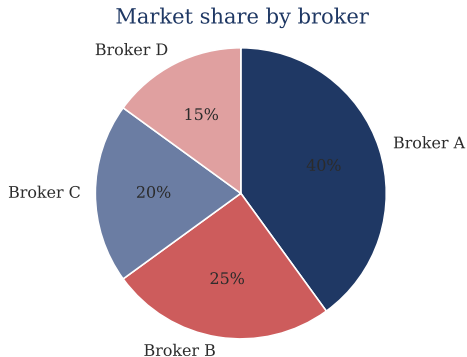
The pie chart

- **Pie chart** (圓餅圖): slices proportional to relative frequency. Relative frequency p gets $360p$ degrees.
- Use it when the point is each category's share of one whole.

Example 2.4

Four brokers split a market with shares 40%, 25%, 20%, 15%. Display the shares.

Example 2.4: pie chart of market share



Each angle is proportional to the category's relative frequency, so we read shares of the whole directly. It works only because the four shares are parts of one total.

The histogram

Definition 2.5 (Histogram)

A *histogram* (直方圖) is the bar graph of the frequency distribution of a numerical variable. The horizontal axis is divided into adjoining class intervals; in this course they always have equal width. Over each interval we draw a bar whose height is the frequency of that interval. Because the intervals adjoin, the bars touch. In a bar chart they do not.

- The intervals are adjacent on the numerical scale, so the picture shows no gap where the scale has none.

Example 2.6

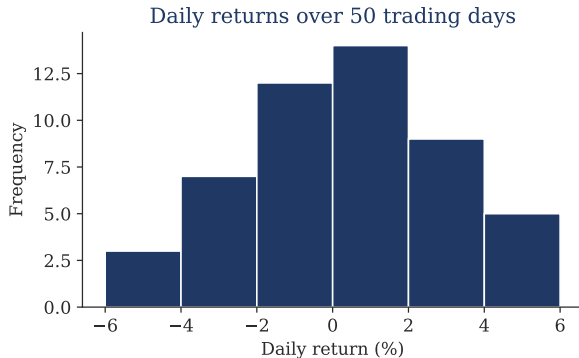
Example 2.6

Fifty daily returns, grouped into six classes of width 2%:

class (%)	$[-6, -4)$	$[-4, -2)$	$[-2, 0)$	$[0, 2)$	$[2, 4)$	$[4, 6)$
frequency	3	7	12	14	9	5

Construct the histogram and describe the distribution.

Example 2.6: histogram of 50 daily returns



It runs from -6% to 6% and is roughly symmetric, centered a little above 0. No single number carries all of that.

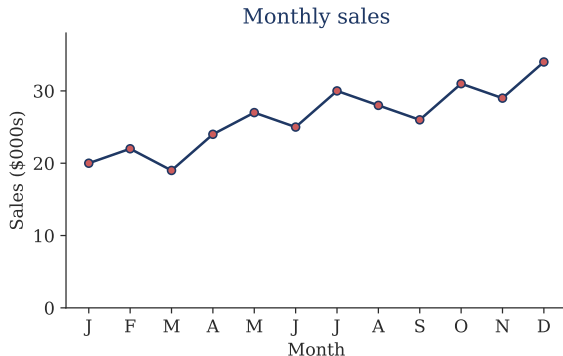
The time-series plot

- **Time-series plot** (時間序列圖): plot the value against time and join the points.
- The order is the information, so the axis must stay in time order.

Example 2.7

Monthly sales (\$000s), January to December:
20, 22, 19, 24, 27, 25, 30, 28, 26, 31, 29, 34.

Example 2.7: monthly sales over a year



Joining the points shows what a bar chart of the same numbers would hide: a clear upward trend, with small ups and downs around it.

The scatter plot

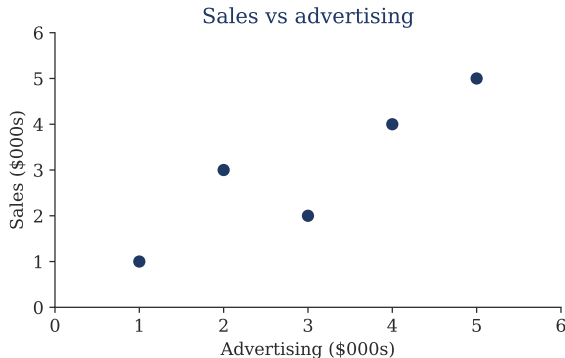
- **Scatter plot** (散布圖): plot each observation as a point (x_i, y_i) . It is the picture behind the correlation of §2.4.

Example 2.8

Five stores report advertising spend x and sales y (\$000s):

$(1, 1)$, $(2, 3)$, $(3, 2)$, $(4, 4)$, $(5, 5)$.

Example 2.8: sales against advertising

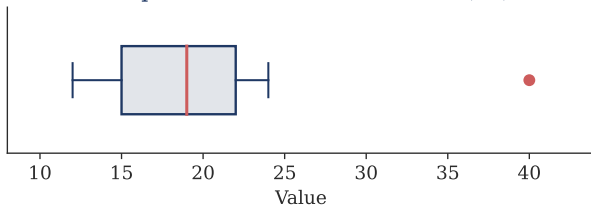


The points rise together from lower left to upper right. §2.4 turns this picture into one number, the correlation $r = 0.90$.

The five-number summary and the box plot

- **Five-number summary:** min, Q_1 , median, Q_3 , max. §2.2 defines the median and the quartiles.
- The **box plot** (盒鬚圖) draws it: box from Q_1 to Q_3 , median inside, whiskers to the most extreme ordinary values.
- A value that lies too far out is drawn as a separate point. §2.3 gives the rule.

Box plot with one extreme value (40)



Example 2.9: the value 40 is drawn on its own; Example 2.26 confirms it.

Measures of Location

Mean, median, mode

Definition 2.10 (Mean, median, mode)

1. *Mean* (平均數): $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, the balance point.
 2. *Median* (中位數): the value in the middle of the ordered data.
 3. *Mode* (眾數): the most frequent value.
- The mean uses every value, so one very large value pulls it up.
 - The median depends only on the central values, so it hardly moves: it is resistant.
 - When they come apart, the gap between mean and median is a useful clue about shape (§2.4). A rule of thumb, not a rule.

Example 2.11

Example 2.11

Ten employees earn (\$000s): 3, 3, 4, 5, 6, 7, 8, 9, 10, 45. Find the mean, median, and mode, and say what their order tells you.

Averaging things that compound

Definition 2.12 (Geometric mean)

The *geometric mean* (幾何平均數) of n positive numbers is

$$\bar{x}_g = \left(\prod_{i=1}^n x_i \right)^{1/n}.$$

For growth, apply it to the factors $(1 + r_i)$.

- Returns and growth rates compound, and the two means answer different questions.
- Geometric: the constant per-period rate reproducing the same ending value.
Arithmetic: the average one-period return.

Example 2.13

Example 2.13

\$100 gains 100% in year 1 (to \$200), then loses 50% (back to \$100). What is the average annual return?

Pieces of different size

Definition 2.14 (Weighted mean)

When x_i carries weight w_i , the *weighted mean* (加權平均數) is

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}.$$

- The ordinary mean is the special case of equal weights.
- Use it whenever the observations stand for pieces of different size: portfolio holdings, course credits, group sizes.

Example 2.15

Example 2.15

A portfolio holds 50% in A (8%), 30% in B (12%), 20% in C (−5%). Find the portfolio return.

Where a value stands in the data

Definition 2.16 (Percentiles, quartiles, five-number summary)

The P th percentile (百分位數) is a value such that about $P\%$ of the observations are at or below it. By the position method it sits in ordered position $\frac{P}{100}(n + 1)$. The quartiles (四分位數) are $Q_1 = P_{25}$, $Q_2 = P_{50}$ (the median), $Q_3 = P_{75}$, at positions $0.25(n + 1)$, $0.50(n + 1)$, $0.75(n + 1)$.

- A percentile reports where a value stands. Only the top tenth of the data exceed the 90th percentile.
- The three quartiles cut the data into quarters, and with the extremes they give the five-number summary.
- **Two conventions:** software (Excel, numpy) uses rank $1 + \frac{P}{100}(n - 1)$. The answers differ, so state which you use.

Example 2.17

Example 2.17

For 4, 6, 7, 8, 10, 10, 11, 12, 13, 14, 15 ($n = 11$) find the quartiles and the five-number summary, then the 60th percentile.

Measures of Variability

Two measures of spread

Definition 2.18 (Range and interquartile range)

The *range* (全距) is $\max - \min$. The *interquartile range* (四分位距) is $\text{IQR} = \underline{Q_3 - Q_1}$, the width of the middle half of the data.

- The range uses only the two extremes, so one extreme value controls it.
- The IQR ignores the outer quarters, so it is resistant. That is why the box plot is built from it.

Spread around the mean

Definition 2.19 (Variance and standard deviation)

For a sample, the *variance* (變異數) and *standard deviation* (標準差) are

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

For a whole population, divide by N instead.

- We square the deviations so that positive and negative ones do not cancel. The root returns to the original units.
- Dividing by n underestimates the population variance on average. Section 7 proves why $n - 1$ corrects it.

Example 2.20

Example 2.20

A fund's monthly returns (%) over five months are 2, 4, 6, 8, 10. Find the range, the sample variance, and the sample standard deviation.

Relative spread

Definition 2.21 (Coefficient of variation)

For $\bar{x} > 0$, the *coefficient of variation* (變異係數) is

$$CV = \frac{s}{\bar{x}},$$

usually reported as a percentage. It is unitless.

- So it compares variability across different units or scales.
- The standard deviation on its own cannot do that.
- It needs a mean that is positive and well away from zero. A mean near zero makes the ratio explode.

Example 2.22

Example 2.22

Asset X: mean 8%, sd 12%. Asset Y: mean 15%, sd 18%. Which is more variable *relative* to its return?

How much data lies near the mean?

Proposition 2.23 (Chebyshev's theorem)

For **any** data set and any $k > 1$, at least $1 - \frac{1}{k^2}$ of the observations lie in $[\bar{x} - ks, \bar{x} + ks]$.

- It assumes nothing about the shape. In return its bound is weak. The proof needs Section 4.
- For approximately normal data, bell-shaped and symmetric, the **Empirical Rule** does better. About 68%, 95% and 99.7% of the data lie within 1, 2 and 3 standard deviations of the mean.

Comparing values across distributions

Definition 2.24 (z-score)

The *z-score* (標準分數) of an observation x is

$$z = \frac{x - \bar{x}}{s}.$$

It counts standard deviations above the mean (positive) or below it (negative).

- It strips away units and scale.
- A z of 2 means “two standard deviations above average”, whether the data are exam scores or returns.

Example 2.25

Example 2.25

Exam scores have mean 72, sd 4. (a) Chebyshev intervals for at least 75% and 88.9%. (b) Empirical Rule for two sd. (c) z-score of 80.

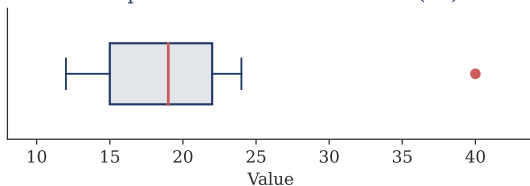
Extreme values and the box plot

- An **outlier** is an observation far from the bulk of the data.
- The standard rule marks off two bounds, called the *fences*:

$$\text{lower fence} = Q_1 - 1.5 \text{ IQR}, \quad \text{upper fence} = Q_3 + 1.5 \text{ IQR}.$$

- Anything outside them is flagged as a *potential* outlier, worth a closer look. The whiskers stop at the most extreme values inside the fences.

Box plot with one extreme value (40)



Example 2.26

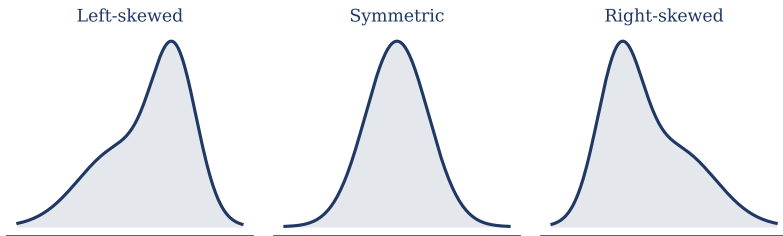
Example 2.26

For 12, 14, 15, 16, 18, 19, 20, 21, 22, 24, 40 find the quartiles and use the fence rule to identify any outliers.



Shape and Association

Shape



- **Skewness** (偏態) is the asymmetry of the two tails: a long right tail is a right skew, a long left tail a left skew.
- This is the picture behind the mean–median comparison of §2.2.
- Kurtosis, how often a distribution produces extreme observations through its tails, is described here. Its formula is deferred to Section 4.3.

Association between two variables

Definition 2.27 (Sample covariance and correlation)

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$
$$r = \frac{s_{xy}}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}, \quad -1 \leq r \leq 1.$$

(Covariance (共變異數); correlation (相關係數).)

- The covariance gives the direction, but its size depends on the units.
- r is unitless and bounded. So it measures the strength of the **linear** relationship.
- r covers linear association only, and correlation is not causation.

Example 2.28

Example 2.28

For $(1, 1), (2, 3), (3, 2), (4, 4), (5, 5)$ (advertising x , sales y) find the sample covariance and correlation.